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Knowledge Backbone

From Copilot discussion

How to Build the Ramboll Knowledge Graph

A practical, architecture-ready blueprint for UPE

1. Start with the Purpose: What the Knowledge Graph is for

Across your meetings and materials, the role of the knowledge graph is consistent:

- A **cross-platform knowledge layer** that sits above ACC, ProjectWise, Trimble, AVEVA, etc. — because none of them will build it for us. [\[DPE Check-...id Thomson | Meeting\]](#)
- A representation of **Ramboll's processes, project types, data objects, relationships and semantics** — the “how we work” layer behind Camino + UPE. [\[DPE Check-...id Thomson | Meeting\]](#)
- A foundation for **AI agents**, retrieval, and decision support. [\[DPE Check-...id Thomson | Meeting\]](#)
- A way to unify **engineering data + documents + metadata + processes**.
- A source to support:

- Q&A over SOPs, instructions, past projects (e.g., M-Files I Know AI examples) [\[Functional...eoptagelse | Video\]](#)
- Engineering domain intelligence (e.g., OCR model → semantic extraction → knowledge layer) [\[TECH | Teams\]](#)
- Cross-project learning and pattern recognition.
- A single “institutional memory” for Ramboll.

So before technical work, define the **Ramboll KG Scope v1.0**:

Entities:

Projects, Clients, Disciplines, Assets, Deliverables, Tasks, Requirements, Risks, Standards, Processes, Tools, People (non-sensitive attributes), Transmittals, Data Models.

Relationships:

(Project) —uses→ (Tool)

(Project) —produces→ (Deliverable)

(Procedure) —applies_to→ (Discipline)

(Document) —belongs_to→ (Project Phase)

(Project) —similar_to→ (Project) (AI-derived semantic similarity)

This gets you 80% of value fast.

2. Build on What Already Exists (Don't start from zero)

A. UPE's Data & Knowledge Layers

Your UPE material already specifies:

- a **data normalization layer**,
- a **knowledge layer**,
- and an **agentic layer** on top. [\[DPE Check-...id Thomson | Meeting\]](#)

The knowledge graph *is* that middle layer.

B. Leverage M-Files as a Document Knowledge Graph Backbone

From the “Functional – M-Files” sessions:

- M-Files explicitly ships with an **enterprise knowledge graph** relating documents ↔ projects ↔ clients ↔ phases. [\[Functional...eoptagelse | Video\]](#)
- It supports **object relationships** (projects, clients, assets as first-class objects).
- It supports **AI-assisted metadata extraction** (I Know).
- It integrates natively with **Microsoft 365 storage**, meaning **Copilot can read the context graph**.
[\[Functional...eoptagelse | Video\]](#)

Even if M-Files is not selected as the DMS, this is a **reference implementation** of what the KG must do.

C. Knowledge Graphs research note (GraphRAG)

Your Knowledge Graphs note from Peter Ewald and Adam Lagoda highlights:

- Microsoft's **GraphRAG** approach
- Issues with **full graph rebuilds** on update

- Need for **incremental/light graph construction**
- Situations where only “local graphs” are needed (i.e., Origami vs RamGPT distinctions) [3]
[https://ramboll.sharepoint.com/sites/AITechnologyDataScienceChapter/_layouts/15/Doc.aspx?action=edit&mobileredirect=true&wdorigin=Sharepoint&DefaultItemOpen=1&sourcedoc={935aa18e-815a-4f64-92de-a606b0aff41f}&wd=target\(/Knowledge Exploration.one/\)&wdpartid={dd0da4ce-7640-3642-93bc-b10f5a71305e}{1}&wdsectionfileid={09e31a65-fce8-49c7-8e61-43c5eb87cb06}\)](https://ramboll.sharepoint.com/sites/AITechnologyDataScienceChapter/_layouts/15/Doc.aspx?action=edit&mobileredirect=true&wdorigin=Sharepoint&DefaultItemOpen=1&sourcedoc={935aa18e-815a-4f64-92de-a606b0aff41f}&wd=target(/Knowledge%20Exploration.one/)&wdpartid={dd0da4ce-7640-3642-93bc-b10f5a71305e}{1}&wdsectionfileid={09e31a65-fce8-49c7-8e61-43c5eb87cb06}))

This is a major design insight and aligns with your need for *scalable, partial-graph* updates.

D. AI Archivist as a KG Use-Case

AI Archivist explicitly lists:

- “contextual knowledge graph insights (v2)” as a target capability [\[AiArchivist | PDF\]](#)

This means:

UPE doesn’t just need a graph — **multiple Ramboll AI products expect one.**

3. Define the Technical Architecture (Minimal Lovable Version)

Layer 1 — Data Ingestion (from UPE Ecosystems)

Sources include:

- Autodesk ACC, ProjectWise, Trimble Connect
- CRM/ERP (project objects)

- SharePoint / Teams
- Document management (M-Files or alternative)
- Camino (process models)
- RamGPT / Origami uploaded content
- Engineering models (via metadata extraction)
- OCR pipelines (DeepSeek VL, GutenOCR etc. mentioned in ML Chapter drops) [\[TECH\]](#) [Teams](#)

Approach:

- Extract only **metadata and relationships** first.
- Do *not* ingest raw files into KG storage — store references.

Layer 2 — Semantic Normalization

You define:

- A unified schema for Projects, Phases, Disciplines, Deliverables, Standards
- Common IDs for projects across platforms (ACC → CRM → ERP)
- A shared taxonomy (building from the UPE production data model) [\[Unified Pr...nvironment\]](#) [PowerPoint](#)

Use lightweight ontologies first (simple JSON-LD schemas).

Layer 3 — Graph Storage

Options:

- Microsoft Fabric OneLake + GraphQL layer
- Azure Cosmos DB (Gremlin)

- Neo4j (vendor-neutral)
- Databricks Lakehouse (since Anubhav discussed Databricks/Mosaic AI as an option for enterprise-scale KG)
[\[DPE Check-...id Thomson | Teams\]](#)

Choose based on where the UPE Data Layer lands.

Layer 4 — Retrieval Layer (GraphRAG)

Use Microsoft's GraphRAG work as your baseline (as highlighted in your internal file).

Supports:

- Semantic chunking
- Subgraph generation on retrieval
- Multi-hop reasoning
- Incremental graph update patterns

Layer 5 — Application Layer

Supports:

- RamGPT enterprise retrieval
- Origami agents
- AI Rule Checker (your meeting with Christoffer B-S) [\[RE: AI Rul...equirement | Outlook\]](#)
- DMS / UPE orchestration
- Process mining & Camino
- Audit & risk reasoning

4. Start With Three High-Value Use Cases (Build the graph iteratively)

Use Case 1 — Cross-Project Knowledge Retrieval

You already described this in your DPE session:

“Have we done similar projects before? Have we designed X in this geography? How did we solve Y?” [\[TECH Teams\]](#)

This is GraphRAG + project metadata + document embedding.

Use Case 2 — Process & SOP Q&A (Quality)

< Person>Darcy Diem's question in the M-Files session is the perfect anchor:

“Can people ask ‘How do I do this?’ and get the SOP?” [\[Functional...eoptagelse | Video\]](#)

This requires:

- SOP → Process → Phase → Deliverable graph
- Retrieval that respects permissions and versioning.

Use Case 3 — Deliverables Relationships & Dependencies

You and Marzia discussed MDR concerns:

- Multiple files belonging to the same document number
- Versioning behaviour
- Transmittal lifecycle [\[Functional...eoptagelse | Video\]](#)

A KG handles this naturally:

- Deliverable node with multiple file nodes
 - Document ↔ Phase ↔ Revision relationships
 - Status transitions linked into graph
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5. Governance — the part most companies skip

You asked in your "Another request for insight vault" email:

"Who owns the definition of data standards, ingestion rules, acceptance criteria?" [\[Re: Anothe...ght vault.\]_Outlook\]](#)

For KG to work:

- DDP (you) owns semantics
- DPE owns platform & backend implementation
- GBAs own domain-specific ontologies
- Camino team owns process graph

- Tech Governance Board approves shared models

Without this: fragmentation.

6. How to Execute (Roadmap)

Phase 0 — 4 weeks

- Define v1 ontology
- Define data sources
- Define identifiers (project, discipline, deliverable)

Phase 1 — 8–10 weeks

- Build the *Knowledge Graph MVP* using:
 - Projects
 - Deliverables
 - SOPs/Processes
 - Documents
 - Basic relationships
- Implement GraphRAG for RamGPT

Phase 2 — 12–16 weeks

- Expand into:
 - Engineering models metadata
 - Transmittals & MDR
 - Cross-platform synchronization (ACC ↔ UPE)

Phase 3 — 2026

- Full semantic UPE Knowledge Layer
- Agentic layer (AI assistants orchestrating flows)
- Unified UX (part of UPE vision)